

Hackathon Project Phases Template that ensures students can complete it efficiently while covering all six phases. The template is structured to capture essential information without being time-consuming.

Hackathon Project Phases Template

Project Title:

Audio Transcription App Using OPENAI Whisper

Team Name:

Visionary Crew

Team Members:

- ❖ Vasudha Nuthalapati
- ❖ Nithisha Pakanati
- ❖ Geetha Supriya Kakaraparthi
- ❖ Divya Kanamarlapudi
- ❖ Bhanu paladugu

Phase-1: Brainstorming & Ideation

Objective:

- Identify the problem statement - Address the challenges of manual transcription, such as time consumption, inaccuracy, and high costs.
- Define the purpose and impact – Develop an AI-powered transcription solution that enhances efficiency, accessibility, and accuracy for users across various industries.

Key Points:

Problem Statement:

Manual transcription is time-consuming, error-prone, and expensive. Many professionals, students, and content creators struggle to efficiently convert speech into accurate text. Additionally, existing solutions often lack affordability, real-time processing, and multilingual

support.

Proposed Solution:

Develop an AI-powered audio transcription app using OpenAI Whisper that provides accurate, real-time, and multi-language speech-to-text conversion. The app will support audio uploads, live transcription, speaker identification, timestamping, and text editing, making it ideal for various use cases.

Target Users:

- Journalists and Podcasters - For transcribing interviews and content
- Students and educators - For taking Lecture Notes
- Business Professionals - For meeting and call Transpirations.
- Content Creators - For generating Captions and subtitles
- Legal and Medical professionals - For case notes and patient Records.

Expected Outcome:

- High-accuracy transcriptions with AI-powered speech recognition.
- Improved productivity by automating transcription tasks.
- Cost-effective and accessible solution for individuals and businesses.
- Enhanced collaboration with easy sharing and exporting options
- Scalability for various industries needing speech-to-text solutions.

Phase-2: Requirement Analysis

Objective: identify key features, system architecture, and dependencies for implementation. Analyze constraints and challenges to ensure feasibility and scalability

Key Points:

Technical Requirements:

- Speech-to-Text Processing – Utilize OpenAI Whisper for accurate transcriptions.
- Multi-Language Support – Transcribe audio in multiple languages.
- Real-Time & Offline Transcription – Support live and pre-recorded audio.
- Speaker Identification – Detect different voices in conversations.
- Timestamps & Formatting – Provide time-based text alignment.
- Audio Enhancement – Noise reduction and clarity improvement.
- Export Options – Save transcriptions as TXT, DOCX, PDF, or SRT.
- Cloud & Local Storage – Store transcripts securely.
- Mobile & Web App – Cross-platform accessibility.
- Integration With third -party services-Google Drive,Dropbox,Notion,etc.

Constraints & Challenges:

- Processing Large Audio Files – Optimize for long recordings without performance issues.
- High Computational Cost – Whisper models require significant GPU resources.
- Privacy & Security Concerns – Ensure compliance with GDPR, HIPAA, etc.
- Accent & Dialect Variability – Handle diverse speech patterns effectively.
- Real-Time Accuracy – Minimize errors in live transcriptions.
- Scalability & Cloud Costs – Balance performance with cost-effective cloud deployment.

Phase-3: Project Design

Objective:



- Design the system architecture and define the user flow.
- Develop a seamless user experience with intuitive UI/UX.
- Ensure scalability and efficiency in handling speech-to-text conversion.

Key Points:

System Architecture Diagram:

➤ A typical architecture for the audio transcription app using OpenAI Whisper would include:

- 1. Frontend (User Interface)
 - Web/Mobile App for user interaction
 - Options to upload or record audio.
 - Display transcription results.
- 2. Backend (Processing Layer)
 - Audio file preprocessing (format conversion, noise reduction).
 - OpenAI Whisper API for speech-to-text conversion.
 - Post-processing (punctuation, speaker identification, timestamping).
- 3. Database & Storage
 - Cloud storage for audio files and transcriptions.
 - User profiles and history management.
- 4. Integration Layer

User Flow:

- 1. User Registration/Login – Authenticate via email, social media, or SSO.
- 2. Upload or Record Audio – Select an audio file or record in real time.
- 3. Processing & Transcription – AI model processes the audio and generates text.
- 4. Text Review & Editing – Users can edit, highlight, and format the transcription.
- 5. Download & Share – Export the final transcript in various formats.
- 6. History & Storage – Access past transcriptions from the dashboard.

UI/UX Considerations:

- Simple & Clean Interface – Minimal design for easy navigation.
- Drag & Drop Upload – Quick access to upload audio files.
- Live Transcription Preview – Show text in real time as audio is processed.
- Dark Mode & Accessibility Options – Improve user experience.

- Multi-Language Support – Switch languages for different users.
- Mobile & Web Optimization – Responsive design for various devices.

Phase-4: Project Planning (Agile Methodologies)

Objective:

- 1. Breaking Down the Tasks Using Agile Methodologies
- Using Agile principles, the transcription process can be divided into structured tasks to ensure efficiency and accuracy.
- 2. Key Points for Agile-Based Audio Transcription Project Planning

Key Points:

Sprint Planning:

- 1. Sprint Planning:
- Identify the core functionalities of the project.
- Break down the project into small, manageable tasks.
- Assign tasks to team members based on their expertise.
- Define the duration of each sprint (e.g., 1-2 weeks).
- 2. Task Allocation
- Developer 1: Backend development (handling API, database, and Whisper integration).
- Developer 2: Frontend development (UI/UX and displaying transcriptions).
- Developer 3: Testing and debugging.
- Project Manager: Overseeing progress and resolving bottlenecks.

Task Allocation:

- Developer 1: Backend development (handling API, database, and Whisper integration).
- Developer 2: Frontend development (UI/UX and displaying transcriptions).

- Developer 3: Testing and debugging.

Timeline & Milestones:

- Week 1-2: Research & setup (Whisper API integration, framework selection).
- Week 3-4: Develop core functionality (audio upload, transcription processing).
- Week 5-6: Implement frontend UI and test API responses.
- Week 7: Final testing and debugging.
- Week 8: Project deployment and documentation.

Phase-5: Project Development

Objective:

- The primary objective of project development in audio transcription using OpenAI Whisper is to create an efficient, accurate, and scalable system for converting spoken language into text. This involves leveraging AI-powered speech recognition while ensuring quality control, adaptability, and seamless integration into various applications

Key Points:

Technology Stack Used:

- Here are the answers for Phase-5: Project Development based on your audio transcription project using OpenAI Whisper:
- Technology Stack Used
- Programming Languages: Python (for backend), JavaScript (for frontend)
- Frameworks: Flask (backend API), React (frontend UI)
- APIs: OpenAI Whisper API (for transcription), Cloud Storage API (for storing audio files)
- Database: Firebase / PostgreSQL (for storing transcriptions)
- Other Tools: Docker (for containerization), GitHub (for version control)

Development Process:

- API endpoint to upload audio files.
- Integrate OpenAI Whisper API for transcription.
- 1. Setup Environment: Install required dependencies (Flask, React, OpenAI API, etc.

Challenges & Fixes:

- API endpoint to upload audio files

Phase-6: Functional & Performance Testing

Objective:

- Ensure the audio transcription app using OpenAI Whisper functions correctly, delivering accurate and efficient results under various conditions.

Key Points:

Test Cases Executed:

- Verify transcription accuracy across different audio qualities, languages, and accents.
- Test for background noise handling and speaker identification (if applicable)
- Check response time and real-time processing capability.

Bug Fixes & Improvements:

- Identify and fix issues like misinterpretation of words, punctuation errors, and formatting inconsistencies.
- Improve handling of long pauses, overlapping speech, and unclear audio segments.

Final Validation:

- Ensure the app meets user requirements, such as ease of use, integration with other platforms (e.g., Google Docs, Notion), and export options.

Deployment (if applicable):

- Host the final version on cloud platforms or mobile app stores.
- Provide a demo link for user testing before full-scale deployment.

Final Submission:

- **Project Report Based on the templates:**
- Document the project scope, objectives, methodology, and results.
- Include details on how OpenAI Whisper was integrated into the transcription app.
- Discuss any challenges and solutions encountered during development.
- **Demo Video (3-5 Minutes):**
- Showcase the working app, demonstrating audio transcription in real-time.

- Highlight key features, such as accuracy, noise handling, and language support.
- Provide an overview of the app's user interface and workflow.
- **GitHub/Code Repository Link:**
- Share a public or private GitHub repository containing the complete project code.
- Include installation steps, dependencies, and usage instructions in the README file
- **4.Presentation:**
- Create a clear and concise slideshow summarizing the project.
- Cover problem statement, technology stack, implementation, and results.
- Add screenshots and live demo snippets to illustrate key functionalities.